MANIPAL UNIVERSITY JAIPUR

Department of Computer and Communications Engineering

<HYPNEXIS> '26

36-HOUR HACKATHON • 24–25 OCTOBER 2026

BUILD / INNOVATE / CREATE

TEAM NAME

Team Kavach

TEAM ID

HXXXX

PROBLEM STATEMENT ID

AI-01

PROBLEM STATEMENT TITLE

Voice-Clone Detection for Phone Scams

THEME / DOMAIN

AI / ML · Cyber Safety

CATEGORY

Software

KAVACH: CATCHING CLONED VOICES

Team Kavach | HXXXX

PROBLEM STATEMENT

- **3–10 s of audio is enough.** A reel, a WhatsApp voice note or a short "wrong number" call gives scammers a voice to copy.
- **Free tools clone it in minutes.** "Papa, accident ho gaya, jaldi paise bhejo" arrives in the real person's voice.
- **Families trust the voice.** Phones give no signal that a voice is synthetic, and instant transfers leave no time to check.
- **Detectors fail in practice.** Most learn one cloner's quirks or dataset shortcuts like silence length, and break on new tools, phone lines and short clips.

HOW THE SCAM WORKS



OUR SOLUTION

- **Kavach** listens to 2–10 s of speech and answers REAL, NOT SURE or FAKE in under a second.
- **Detects the vocoder fingerprint,** not the tool: every cloner ends in a vocoder, so it keeps working on cloners it has never seen.
- **Built for real calls:** phone codecs, noise, live streaming every 0.5 s and the exact faked seconds inside a call.



TECHNICAL APPROACH

Team Kavach | HXXXX

SYSTEM ARCHITECTURE

1 Clean	trim silence, equal loudness
2 Phone	8 kHz line, AMR / Opus codec, noise (training)
3 Ears: XLS-R 300M	frozen, pretrained on 436k h, 128 languages
4 Mixer	learned weights over all 25 layers
5 Judge	small MLP → score 0–100
6 Verdict	REAL / NOT SURE / FAKE + 0.5 s timeline

CORE TECHNOLOGIES

- **Model:** PyTorch, XLS-R 300M + layer-mix MLP
- **Data:** In-the-Wild, LibriSpeech, edge-tts + Kokoro TTS
- **App:** FastAPI + WebSocket, ffmpeg, Demucs
- **Runs on:** RTX 4050 laptop GPU, or CPU

UNIQUE POINTS

- **Frozen backbone:** only a small head is trained, so it cannot overfit one cloner
- **Shortcut-proof:** silence and loudness removed; a shortcut-only model scores 47% EER (coin toss)
- **Honest output:** NOT SURE instead of a guess; partial-fake localisation

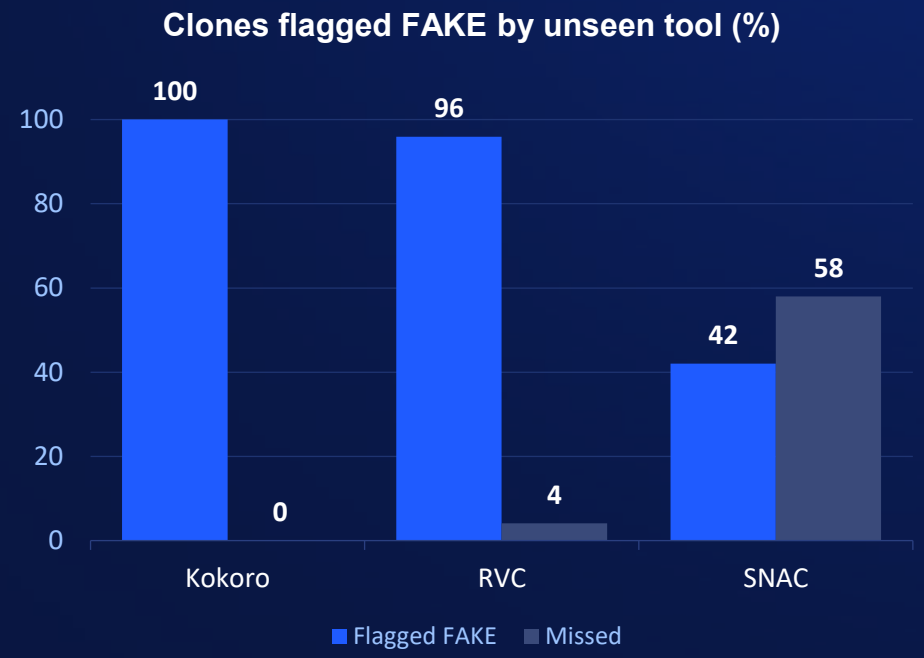
RESULTS ON UNSEEN DATA

Team Kavach | HXXXX

TESTED ONLY ON SPEAKERS AND CLONING TOOLS IT NEVER SAW

Test (never in training)	EER	Flagged FAKE	n
In-the-Wild, unseen speakers, 4 s	2.0%	97.1%	4026
Same, 2 s clips	2.3%	96.8%	4026
Same, phone-line audio	3.0%	95.0%	4026
Unseen modern TTS (MS neural, Kokoro)	0.5%	98.0%	204
Same, replayed speaker → mic	10.3%	93.1%	204
Unseen cloner: RVC voice conversion	1.0%	95.9%	97
Unseen cloner: SNAC neural codec	8.7%	42.0%	300

EER = equal error rate (lower is better, 50% = coin toss). Every number comes from scripts in the repo (runs/.json).*

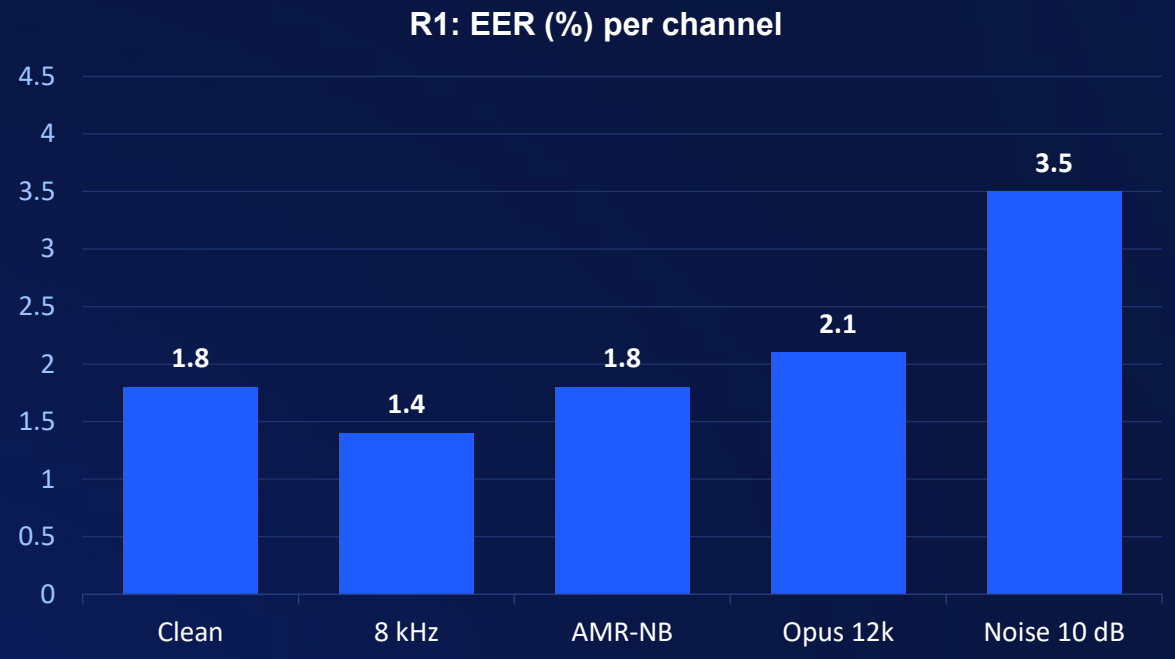


Weak spot we report: SNAC codec clones (vocoder-only fingerprint).

PHASE 2: REAL CALL CONDITIONS

Team Kavach | HXXXX

R1 CODECS & NOISE · R2 PARTIAL FAKES · R3 LIVE STREAM · R4 PHASE 1 HOLDS



Worst case noise = 1.94× clean EER (target ≤ 2.5×). Real ffmpeg encoders.

80.5%

R2 partial-fake F1

faked 1–3 s found at 0.5 s resolution
precision 79.9% · recall 81.0%

500 ms

R3 live call updates

WebSocket stream, 25.7 ms median compute
p95 31.4 ms per update

≤ 0.3 pts

R4 Phase 1 EER change

all Phase 1 tests within 2 points
unseen speakers 2.3% → 2.0%

USER EXPERIENCE & DESIGN

Team Kavach | HXXXX

USER INTERFACE

- One big verdict dial in green, amber or red
- Upload, record or live-call mode
- Timeline marks the faked seconds
- Large type, colour plus words

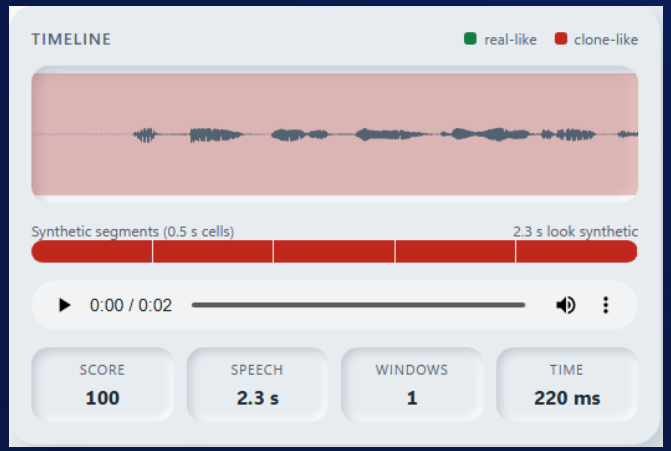


USER FLOW

1. Upload or record a clip, or start a live call
2. Score updates every 0.5 s; faked seconds turn red
3. Clear advice: "Hang up, call back on a saved number"

DESIGN GOAL

A parent with no tech background should know in one second whether to trust the voice, and what to do next.



FEASIBILITY AND VIABILITY

Team Kavach | HXXXX

TECHNICAL FEASIBILITY

- **Working prototype** today: upload, record and live-call modes
- **Speed:** median 318 ms per clip, 0 errors in 60 concurrent requests
- **Tested:** 53 unit / API / streaming / security tests + 13/13 browser end-to-end
- **Modular:** swap backbone or head without touching the app

318 ms

median time to a verdict

ECONOMIC FEASIBILITY

- **Prototype cost:** ₹0 in licences, open-source stack, one laptop GPU
- **Scaling:** frozen backbone, one GPU serves many calls; CPU fallback
- **Hardware guard:** ~₹1,300 Bluetooth device for parents' phones
- **Revenue:** per-call API for banks, telecom licence

₹0

licence cost, open-source stack

OPERATIONAL VIABILITY

- **Deploy:** on-premise API, browser app, later phone app
- **Risk:** new cloners. Fix: retrain head only, held-out tool tests
- **Risk:** replay false alarms (~19%). Fix: NOT SURE band + noise shield
- **Privacy:** audio in memory only, never stored or logged

0

audio files stored or logged

IMPACT AND BENEFITS

1 SOCIAL IMPACT

Protects families and elderly people from "emergency" voice scams

Works on Indian phone audio, Hindi and Hinglish voices

2 PRIVACY & EFFICIENCY

Only a small head is trained: low compute, no huge training runs

Runs locally; no audio sent to or stored in the cloud

3 TECHNICAL IMPACT

Generalises to unseen cloners via frozen speech foundation model

Localises the faked seconds inside a real call (F1 80.5%)

4 EDUCATIONAL IMPACT

Shows people that a familiar voice is no longer proof

Open, reproducible results with weak spots reported

BUSINESS MODEL & SUSTAINABILITY

Team Kavach | HXXXX

TARGET USERS

- **Families:** parents and elderly on phone calls
- **Banks & call centres:** verify callers who claim to be customers
- **Telecoms:** flag suspicious calls on the network
- **Early adopters:** fintech, cyber-cell awareness drives

3 segments

families · banks · telecoms

REVENUE & COST MODEL

- **Pilot:** free web app + campus / cyber-cell demo
- **B2B API:** pay per verified call or monthly SaaS
- **Consumer:** free app; paid Bluetooth guard device
- **Cost:** GPU hosting; head retrains are cheap

Free + B2B

free for families, paid per call for business

SUSTAINABILITY

- **Partners:** banks, telecoms, cyber police
- **Next:** more Indian languages, scam-phrase detection
- **Keep up:** add each new cloner as a held-out test
- **Open benchmark** of Indian-language clone calls

Head only

cheap retrain when new cloners appear

RESEARCH AND REFERENCES

KEY TECH & RESEARCH SOURCES

- **XLS-R:** Babu et al., "XLS-R: Self-supervised cross-lingual speech representation learning at scale", arXiv:2111.09296, 2021
- **In-the-Wild:** Müller et al., "Does Audio Deepfake Detection Generalize?", Interspeech 2022
- **SSL + augmentation:** Tak et al., "Automatic speaker verification spoofing and deepfake detection using wav2vec 2.0 and data augmentation", Odyssey 2022
- **ASVspoof:** Wang et al., "ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech", 2020
- **LibriSpeech:** Panayotov et al., ICASSP 2015 (real speech)
- **Cloners tested:** Kokoro TTS, Microsoft edge-tts, RVC, SNAC (Siuzdak 2024), F5-TTS (Chen et al. 2024)
- **Tools:** PyTorch, Hugging Face Transformers, FastAPI, ffmpeg, Demucs (Défossez et al.)

HYPNEXIS '26

KAVACH

Catching AI-cloned voices
before the money moves.

REAL · NOT SURE · FAKE

THANK YOU

Team Kavach | HXXXX | AI-01 Voice-Clone Detection

CREATE / AN / IMPACT